Ojas Shrikant Madghe

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.78
Aditya English Medium School	Class XII (CBSE)	2023	92.40%
D.A.V Public School Aundh Pune	Class X (CBSE)	2021	98.20%

EXPERIENCE

Nutanix

MTS Intern (Member of Technical Staff)

- Architected an Agentic Framework for cloud-native systems to automate system testing and accelerate production release cycles.
- Designed and implemented modular agent orchestration components using agentic AI frameworks and cloud-native stacks.
- Researched deterministic-first AI application development, evaluating hybrid architectures where workflows selectively invoke LLMs
- Deepened domain expertise in cloud-native infrastructure through Kubernetes training and hands-on container orchestration.
- Collaborated with engineering teams during Intern Hackathon, driving product ideation, technical architecture, and final presentation.

Intelligent Transport Systems Lab- Kyoto University

Research Intern

- Engineered a physics-aware model processing Probe Request data from Wi-Fi sensors for privacy-compliant population estimation.
- Utilized SciPy to learn non-linear physics parameters and benchmarked against traditional ML algorithms (e.g., Linear Regression).
- Contributed to constructing standardized training datasets from large-scale raw Wi-Fi sensor feeds.
- Presented research findings at the KU-STAR Poster Presentation event, earning strong positive feedback.

Invention Factory 2024, Maker Bhavan Foundation

Inventor (Intern)

- Selected from over 750 national applicants for an intensive program focused on prototyping, engineering design, and IP strategy.
- Co-invented and filed provisional patent applications in the U.S. and India for a novel Band-Aid Applicator Device

ACHIEVEMENTS

- Awarded the Academic Excellence Award for Highest GPA in the department
- Solved 200+ problems across coding platforms (LeetCode, GeeksForGeeks, Coding Ninjas)

SKILLS

C/C++, Python, SQL, Linux/Unix, Git, Flask, NumPy, Pandas, Matplotlib, MySQL, MongoDB, Kubernetes, Docker, Microservices, Agentic AI, TensorFlow, Keras, Scikit-Learn, SciPy, LLM Orchestration.

RELEVANT COURSES

Computer Architecture, Introduction to Programming, Introduction to Modern AI, Probability, Statistics, Discrete Mathematics, Linear Algebra, Data Structures & Applications, Algorithms, Theory of Computation, Foundations of Machine Learning

PROJECTS

KVM-Hypervisor Observability Platform (Python)

- Designed a modular metrics pipeline (collector, aggregator, API, dashboard) that polls host and guest VM performance data at fixed intervals using raw /proc filesystem reads and virsh
- Computed derived system health metrics including CPU utilization, disk I/O wait, memory availability, swap/page-fault growth rates, and per-VM resource-fairness scores from scheduler statistics
- Built a rule-based alerting layer using short- vs. long-window trend comparisons to detect rising memory pressure ahead of hard thresholds
- Served real-time state via a FastAPI backend and visualized it in an auto-refreshing Streamlit dashboard

Road Damage Classifier using CNNs

- Developed a computer vision model to classify road conditions (damaged / undamaged) using image datasets.
- Employed data augmentation, transfer learning (Keras/TensorFlow); hosted on a Flask-based web interface.
- Gained foundational CNN knowledge via the deeplearning.ai Convolutional Neural Networks course.

University Admission Predictor

- Designed a machine learning-based web app to predict graduate school admission chances using CGPA, GRE, TOEFL, and research experience.
- Applied EDA, feature engineering, regression models (Scikit-learn); deployed using Flask.

EXTRACURRICULAR

- Mercedes-Benz SafeRoads Ideathon (Institute Finalist)
- Global Young Entrepreneur Competition (hosted by the Entrepreneurship Faculty of Musashino Campus of Tokyo) (Institute Finalist)
- Smart India Hackathon